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Development of a 3D cell printed construct considering angiogenesis for liver tissue engineering

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Abstract

Several studies have focused on the regeneration of liver tissue in a two-dimensional (2D) planar environment, whereas actual liver tissue is three-dimensional (3D). Cell printing technology has been successfully utilized for building 3D structures; however, the poor mechanical properties of cell-laden hydrogels are a major concern. Here, we demonstrate the printing of a 3D cell-laden construct and its application to liver tissue engineering using 3D cell printing technology through a multi-head tissue/organ building system. Polycaprolactone (PCL) was used as a framework material because of its excellent mechanical properties. Collagen bioink containing three different types of cells—hepatocytes (HCs), human umbilical vein endothelial cells, and human lung fibroblasts—was infused into the canals of a PCL framework to induce the formation of capillary-like networks and liver cell growth. A co-cultured 3D microenvironment of the three types of cells was successfully established and maintained. The vascular formation and functional abilities of HCs (i.e., albumin secretion and urea synthesis) demonstrated that the heterotypic interaction among HCs and nonparenchymal cells increased the survivability and functionality of HCs within the collagen gel. Therefore, our results demonstrate the prospect of using cell printing technology for the creation of heterotypic cellular interaction within a structure for liver tissue engineering.

1. Introduction

The liver performs various essential functions for our survival, such as the control of plasma glucose levels, the synthesis of various proteins, including albumin and coagulation proteins, the synthesis of bile for the absorption of fat in the digestion process, detoxification by transaminase reactions, and the storage of several vitamins [1]. In the United States, 2.5 million patients suffer from end-stage liver disease (ESLD), and more than 25 000 die from it each year [2]. Current treatment options for ESLD are limited to liver transplantation, and most patients die because of the shortage of donor organs

[3]. Many researchers have focused on alternative treatments, such as primary hepatocyte (HC) cell implantation systems, to overcome the organ shortage, because HC transplantation has advantages compared with whole-liver transplantation. The required number of primary HCs can be obtained from a single donor liver, which facilitates the treatment of several patients. The isolated primary HCs can be cryopreserved, enabling them to be used later [4]. The primary HCs are also incorporated into a biodegradable scaffold to enhance the survival and functional stability of the implantation [5, 6]. Therefore, tissue engineering is drawing increasing interest for addressing the shortage of tissues and organs.

Tissue engineering aims to recover the original function of impaired tissues by combining knowledge of engineering and life sciences to replace damaged tissues with new tissues or regenerated organs. In the future, engineered liver tissue will improve cell replacement, regenerative medicine, and the design of *ex vivo* platforms for the development of better antivirals and the evaluation of the hepatotoxicity of new drug molecules [7, 8]. However, developing engineered liver tissue is challenging, because isolated HCs die or lose their original functions due to the loss of interaction with the extracellular matrix (ECM) and surrounding cells and the disruption of their normal microenvironment [9]. Therefore, to recover the function of the liver, the native microenvironment must be achieved [10, 11]. To enhance the functionality of HCs, recent studies have reported the co-cultures of HCs, fibroblasts, and human umbilical vein endothelial cells (HUVECs) and the use of two-dimensional (2D) patterning to induce interaction among cells [3, 12]. However, these studies were limited to the 2D environment, whereas actual liver tissue is a three-dimensional (3D) environment.

To address the limitations of 2D cell cultures, several 3D culture approaches have been explored. One such approach involves coating the culture dish with ECM molecules [13, 14]. Various HC culture systems, including hollow fibers and macroporous polymer scaffolds, have been developed and used to evaluate liver-specific functions, such as albumin secretion and detoxification capacity [15, 16]. Hepatocyte encapsulation using a heparin-based hydrogel and 3D cell culture with hepatic cell sheets has been reported [17–19]. However, these techniques cannot be used for efficiently building large-volume 3D complex constructs for surgical implantation.

Recently, cell printing technology has been demonstrated as an effective method for printing 3D tissue constructs [20–22]. This technology is a type of 3D printing that uses computer-aided design (CAD) and computer-aided manufacturing (CAM) [23, 24], and it can effectively build 3D microenvironments for target tissues by printing various cells in predetermined positions. Furthermore, the 3D printing of cell-laden constructs can address the problems of insufficient cell seeding and/or non-uniform cell distribution that are associated with manual cell seeding over solid freeform-fabricated scaffolds [25, 26]. This is performed by printing cells uniformly throughout the constructs with a higher cell density while encapsulating them in a hydrogel, in which homotypic or heterotypic cell adhesion or specific microenvironment conditions can be established similar to those of native tissue in the 3D context [27]. In the cell printing process, various materials can be printed including biocompatible materials, cells, and supporting components within one construct by using multiple heads. The cell is encapsulated within a hydrogel, and various kinds of hydrogels can be selected based on the target

tissue [21, 28]. Hydrogels serve the purpose of ECMs, which assist in cell adhesion, proliferation, and differentiation by providing physical and chemical instructive signals to the cells. Various hydrogels, either alone or in combination (e.g., gelatin, chitosan, alginate, poly[ethylene glycol] diacrylate [PEGDA]), along with some ECM components have been exploited for liver tissue engineering, but they are not sufficiently strong for maintaining structural integrity [29, 30]. Thus, having a supporting framework made of solid biomaterial with excellent mechanical properties should increase the stability of the 3D microenvironment.

In this study, we developed 3D printed constructs for liver tissue engineering by using a synthetic polymer framework and a natural polymer-based hydrogel providing a co-culture system for primary rat HCs, HUVECs, and human lung fibroblasts (HLFs). We hypothesize that the heterotypic interaction between the HCs and nonparenchymal cells, provision of signals by the ECM, and sustainment of the 3D microenvironment by the supporting biomaterial will increase the survivability and functionality of HCs in engineered liver tissue.

2. Materials and methods

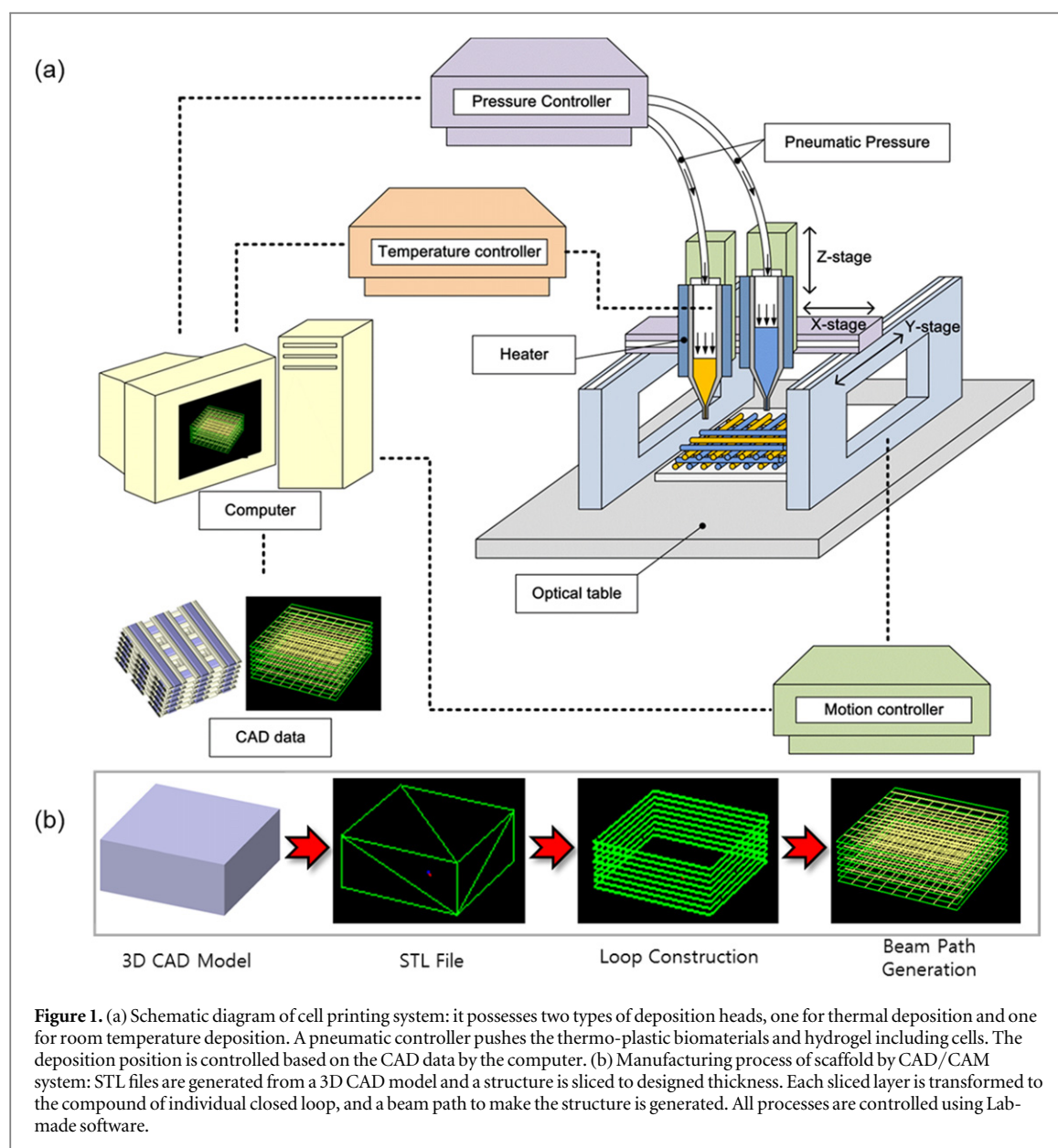
2.1. Cell cultures

2.1.1. Primary rat hepatocytes (HCs)

Primary HCs were isolated from 5 week old female Sprague Dawley rats (Koatech, Busan, Korea) weighing from 140–160 g using a modified two-step collagenase perfusion procedure [31]. The HC viability was estimated to be from 92% to 99% by a trypan blue exclusion test. The HC culture medium consisted of Dulbecco's Modified Eagle's medium (DMEM) supplemented with 10% (v/v) fetal bovine serum (FBS: Gibco, Gaithersburg, MD, USA), 0.5 U ml^{-1} of insulin (Sigma-Aldrich, St. Louis, MO, USA), 7 ng ml^{-1} of glucagon (Sigma-Aldrich), 20 ng ml^{-1} of epidermal growth factor (EGF: Sigma-Aldrich), $7.5 \mu\text{g ml}^{-1}$ of hydrocortisone (Sigma-Aldrich), 200 U ml^{-1} of penicillin, and $200 \mu\text{g ml}^{-1}$ of streptomycin (Gibco BRL, Grand Island, NY, USA). The culture medium was changed daily. All of the animal studies were approved by the Institutional Animal Care and Use Committee at POSTECH Biotech Center, Pohang University of Science and Technology (Permit number: 2013-01-0015).

2.1.2. Human endothelial cells

HUVECs (a gift from Dr Yongsong Gho at POSTECH) were cultured in an Endothelial Cell Growth Medium-2 BulletKit (EGM2 BulletKit, CC-3162; Lonza, Allendale, NJ, USA) in 75 cm^2 tissue culture flasks (BD Biosciences, Bedford, MA, USA). The cells were passaged every 3–4 d, and the medium was changed daily. Cells between Passages 3 and 6 were used in the



experiment. The cells were kept in a standard cell culture incubator at 37 °C with 5% CO₂.

2.1.3. Human lung fibroblasts (HLFs)

HLFs were purchased from Lonza (USA). They were expanded in DMEM supplemented with 10% (v/v) FBS, 2 mM of L-glutamine, 50 U ml⁻¹ of penicillin, and 50 µg ml⁻¹ of streptomycin (Gibco BRL, Grand Island, NY, USA) and kept in an incubator at 37 °C with 5% CO₂.

2.2. 3D cell printed construct preparation

2.2.1. 3D printing system for fabrication of 3D scaffold

We used an in-house-developed multi-head tissue/organ building system (MtoBS) (figure 1(a)) based on 3D printing technology to fabricate the 3D scaffold [32]. This system was designed to print various solid biomaterials and hydrogel-based biomaterials by utilizing pneumatic pressure. The system possesses

four individual deposition heads that can move independently. Each head contains a heating source that is linked to a heating controller, and the heat source can melt thermo-plastic biomaterials, such as polycaprolactone (PCL) and polylactic glycolic acid. Molten materials are deposited by pneumatic pressure and then cooled quickly at room temperature (RT). The heating temperature and pneumatic pressure can be raised to 150 °C and 650 kPa, respectively. A linear motor, linear encoder, and guide were installed to control x-axis and y-axis motions with an accuracy of 2 µm and repeatability of 1 µm. Using this system, the PCL line width was adjustable from 50 to 275 µm by varying the nozzle diameter, pressure, and feed rate [21]. In the z-axis, the accuracy and repeatability were both 5 µm. The working space of this system was designed to be 180 mm × 180 mm × 50 mm to allow fabrication of relatively large volume constructs.

Table 1. Contents of treatment Groups 1–3.

Component	Group		
	1	2	3
HC (cell ml ⁻¹)	1.0×10^6	2.5×10^5	2×10^5
HLF (cell ml ⁻¹)	0	7.5×10^5	6×10^5
HUVEC (cell ml ⁻¹)	0	0	2×10^5

2.2.2. Preparation of cell-laden construct

To observe the angiogenesis of HUVECs in the hydrogel, we encapsulated HUVECs and HLFs in 2% neutralized solution-type atelocollagen (Koken, Tokyo, Japan). The HUVECs and HLFs were mixed at a ratio of 1:3 in the hydrogel, and the total density of the two cell types was set to 1×10^5 cells ml⁻¹ (i.e., 2.5×10^4 HUVECs and 7.5×10^4 HLFs were encapsulated in 1 ml of collagen solution). EGM2 BulletKit was added to the hydrogel at a concentration of 5% (v/v). The cell-laden hydrogel was printed into a single-layer 3D cell printed construct.

To investigate the viability of the HCs in various *in vitro* conditions, we used 2% atelocollagen as the encapsulation material. Three experimental groups were designed, all of which contained 1×10^6 cells ml⁻¹ with 10% HC culture medium and FBS encapsulated in hydrogel, but differed in the types and concentrations of the cells used (table 1). Group 1 included only HCs, Group 2 included a 1:3 mixture of HCs and HLFs, and Group 3 contained a 1:3:1 mixture of HCs, HLFs, and HUVECs. The cell-laden hydrogel was printed into a multi-layer 3D cell printed construct.

2.2.3. Fabrication of 3D cell printed construct

The MtoBS system uses CAD/CAM technology to produce 3D constructs by stacking 2D patterns in the direction of the z-axis (figure 1(b)) [33, 34]. Therefore, information on the 2D pattern is essential for creating the 3D scaffold. We created the 2D pattern by slicing the STL file data extracted from a solid model to generate closed loops. The collected 2D pattern information for all of the layers was loaded into the cell printing system. Finally, with this pattern, we could build the 3D construct as designed.

Using these procedures, we fabricated a 3D printed construct from a synthetic biomaterial and collagen solution using MtoBS (figure 2(a)). First, as a synthetic biomaterial, PCL (molecular weight = 65 000 g mol⁻¹; Sigma-Aldrich, St. Louis, MO, USA) patterns were printed at 80 °C with a metal nozzle with a diameter of 250 µm. A pneumatic pressure of 300 kPa with a feed rate of 100 mm min⁻¹ was applied to fabricate the structure. The line width of the pattern was 200 µm, and the pitch distance was 1 mm. The stacking sequence was 0°/90°, and each pattern was deposited three times with a height of 100 µm. The pattern was deposited three times with a height of 100 µm. The surrounding temperature of the MtoBS

was maintained at 18 °C to cool the PCL pattern down rapidly. Consequently, we successfully fabricated a framework with the overall dimensions of 10.2 mm × 10.2 mm × 0.3 mm. After fabrication of the framework, the atelocollagen encapsulating the HUVECs and HLFs was deposited in the space between the lines. The nozzle end was located in the middle of the space placed between the lines using CAD/CAM-based software to infuse the hydrogel solution into the scaffold. The cell-encapsulated collagen solution was deposited by pneumatic pressure in the scaffold. The same process was repeated to produce a 3D cell printed construct. This atelocollagen mixture was gelled following 30 min of incubation at 37 °C in a 5% CO₂ atmosphere. The cell printed construct was then supplied EGM-2 supplemented with 10% (v/v) FBS and 1% (v/v) penicillin and cultured in a static culture environment for 14 days.

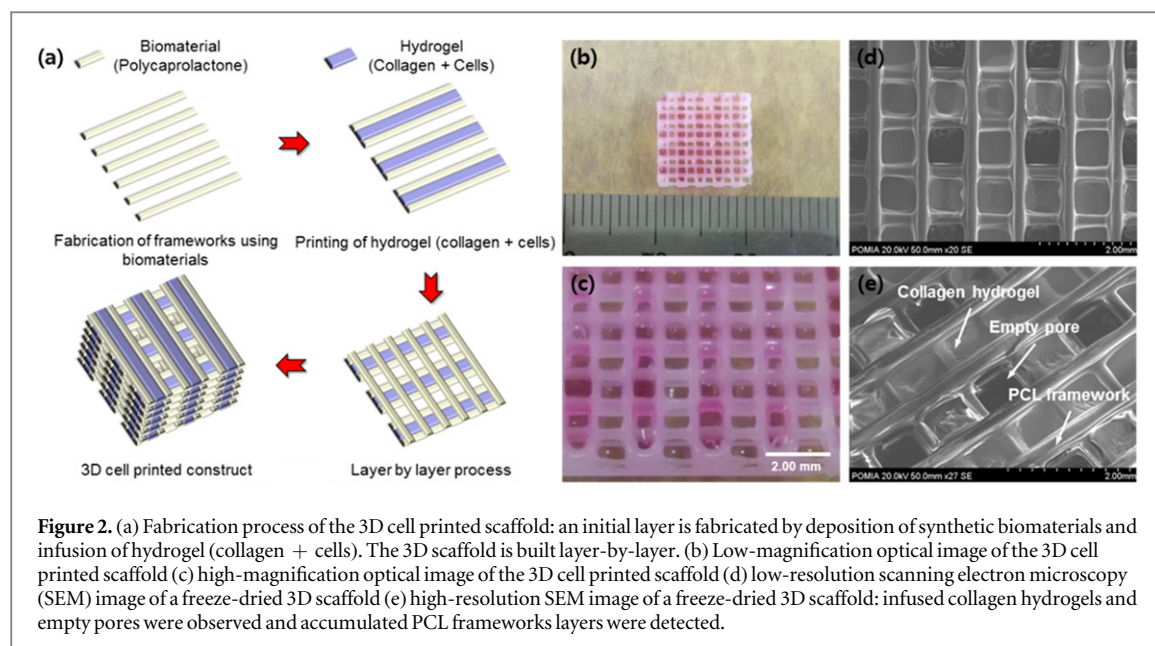
2.3. Cell staining for confocal microscopy

2.3.1. CD31 and lectin staining

The 3D cell printed constructs were fixed in 5% (v/v) formalin for 30 min, washed with phosphate-buffered saline (PBS) three times, and then treated with 0.1% Triton X-100 in PBS for 10 min. The resulting samples were washed with PBS several times and blocked with 1% (v/v) bovine serum albumin (BSA; Sigma-Aldrich, St. Louis, MO, USA) in PBS for 1 h at RT. After washing, the samples were incubated with a 1:50 dilution (v/v) of rabbit anti-CD31 polyclonal antibodies (Abcam, Cambridge, MA, USA) in PBS with 1% (v/v) BSA overnight at 4 °C, and then incubated with a 1:200 dilution of goat anti-rabbit IgG conjugated to Alexa 488 (Molecular Probes, Eugene, OR, USA) and 10 µg ml⁻¹ of BSL (lectin from *Bandeiraea simplicifolia* TRITC conjugate, Sigma Aldrich) in PBS for 1 h at RT. The samples were then washed with PBS three times and mounted using fluorescence mounting medium with 4',6-diamidino-2-phenylindole (DAPI) (GBI-Golden Bridge International, Mukilteo, WA, USA). Finally, the samples were photographed through a FluoView 1000 confocal microscope (Olympus, Tokyo, Japan). All images were captured using the same camera settings.

2.3.2. Albumin staining

The sample preparation steps, from fixation to blocking, were the same as those of the CD31 staining protocol. A 1:500 dilution of sheep anti-rat albumin antibody (Bethyl Laboratories, Inc., Montgomery, TX, USA) in PBS was prepared and then added to the samples. The samples were incubated overnight at 4 °C, washed three times in PBS, and then incubated with a 1:200 (v/v) dilution of donkey anti-sheep IgG conjugated to Alexa 488 (Molecular Probes, Eugene, OR, USA) for 1 h at RT. After washing, the samples were mounted in fluorescence mounting medium with DAPI and photographed through a confocal



microscope (TCS SP5, Leica Microsystems Ltd, Germany).

2.3.3. Analysis of tubular structures

A morphometric analysis was performed on the images acquired through the confocal microscope, which displayed the distribution and growth of HUVECs inside the collagen hydrogel. The sagittal-sectional area of tube-like structures was measured and averaged using ImageJ software (National Institutes of Health, USA) 1, 3, 5, 7, 10, and 14 d after printing.

2.4. Morphology observation using scanning electron microscopy (SEM)

The morphology of the 3D cell printed construct was examined using a SEM (Hitachi-SU6600, Hitachi, Tokyo, Japan) at 15 kV. All samples were coated with titanium using a sputter coater (Ion Sputter E-1045, Hitachi, Tokyo, Japan) for 120 s.

2.5. Biochemical assay

2.5.1. Assay for albumin secretion

The albumin secretion of human HCs in the co-culture systems was evaluated at Days 1, 4, 7, and 10 of culture time. Samples from the culture medium were collected in pre-chilled tubes and stored at -20°C until assayed. The albumin secretion was measured in the samples using the immunoenzymatic ELISA method; 96-well plates were coated with $100\ \mu\text{l}$ of diluted (1:100) coating antibody and left for 1 h at RT. After washing five times in PBS, $200\ \mu\text{l}$ of blocking solution (0.14 M of NaCl and 1% (v/v) of BSA dissolved in 50 mM of Tris buffer) was added to the well plates, and the solution was left for 30 min. After washing five times in PBS, $100\ \mu\text{l}$ of cell culture

supernatant was added to the wells and incubated for 1 h at RT with $100\ \mu\text{l}$ of anti-human albumin monoclonal antibody conjugated with horseradish peroxidase (Bethyl Laboratories, Inc., Montgomery, TX, USA). The samples were then washed five times in PBS, the substrate buffer containing tetramethylbenzidine (Thermo Scientific, Asheville, NC, USA) was added and left for 15 min, and the reaction was stopped using $100\ \mu\text{l}$ of 0.18 M H_2SO_4 . The absorbance was measured at a wavelength of 450 nm using a microplate reader (UVM 340, Biochrom Ltd, Cambridge, UK).

2.5.2. Assay for urea synthesis

The urea synthesis of human HCs in co-culture systems was evaluated at Days 1, 4, 7, and 10 of culture time. Samples from the culture medium were collected in pre-chilled tubes and stored at -20°C until assayed. The urea concentration was determined using the Stanbio Urea Nitrogen Kit (Stanbio Laboratory, Boerne, TX, USA). Standard curves were generated using purified rat urea supplied from a test kit. The absorbance was measured at a wavelength of 600 nm using a microplate reader (UVM 340, Biochrom Ltd, Cambridge, UK).

2.6. Statistical analysis

Data from several separate experiments were expressed as means $\pm$ standard deviation. Data were analyzed using linear regression and one-way analysis of variance (MINITAB version 14.2, Minitab, State College, PA). Values of $P < 0.05$ were considered statistically significant.

3. Results and discussion

3.1. 3D cell printed structure fabrication results

To observe angiogenesis in the 3D cell printed construct, we prepared a single-layer 3D cell printed construct with HUVECs and HLFs in a collagen hydrogel. In this study, we chose the 3D lattice geometry to show the potential of the cell printing technique, which is based on the combination of a polymer framework and hydrogel-encapsulating cells. Throughout the 14 d of HUVEC culture time, the hydrogel sustained its original shape and remained attached to the PCL framework without degradation. The collagen gel (2%) itself is very weak and can easily collapse. The PCL framework protects the collagen gel from collapsing and maintains the shape of the construct. These results demonstrate the feasibility of printing hybrid constructs using our 3D cell printing system.

Using the same conditions as those for the single-layer 3D cell printed construct, we fabricated a four-layer 3D cell printed construct using a cell-laden hydrogel and PCL (figure 2(a)). The final size of the 3D cell printed construct was 10.2 mm × 10.2 mm × 1.2 mm. We could also fabricate the small-scale liver-shape construct through the same process (figure S2). This process successfully produced a 3D cell printed construct with a co-culture system of HCs, HUVECs, and HLFs (figure 2(a)). Furthermore, the collagen hydrogel was intentionally infused into every second canal to create porous 3D cell printed constructs. This result can be achieved only through deposition-based 3D printing technology, because immersion or injection methods cannot intentionally control the position of the hydrogel solution within the 3D cell printed construct. The intentional control of hydrogel infusion within cavities resulted in a fully interconnected porous 3D cell printed construct. Cells entrapped in hydrogel may necrotize due to the limitation in the depth of the mass transport of oxygen, nutrients, and waste by diffusion [35]. However, in our earlier study, we evaluated the cell viability within this kind of lattice structure and found that it was more than 90% soon after printing as well as after 14 d of culture [20, 21]. Hence, we believe that the cells encapsulated in our hydrogel will remain alive for an extensive period, because the nutrients and oxygen can pass through the interconnected pores. In addition, the PCL framework, which possesses excellent mechanical properties, provides the 3D cell printed construct with long-term stability without allowing leakage of the hydrogel. Although the PCL framework could successfully act as a supporting material, it is rather stiff and has a slow degradation rate, and thus, it may not be easy to use for clinical applications. Currently, a bioink necessary for cell printing technology is being developed. The bioink should have printability, shape fidelity, biocompatibility, degradation kinetics, strong structural and mechanical

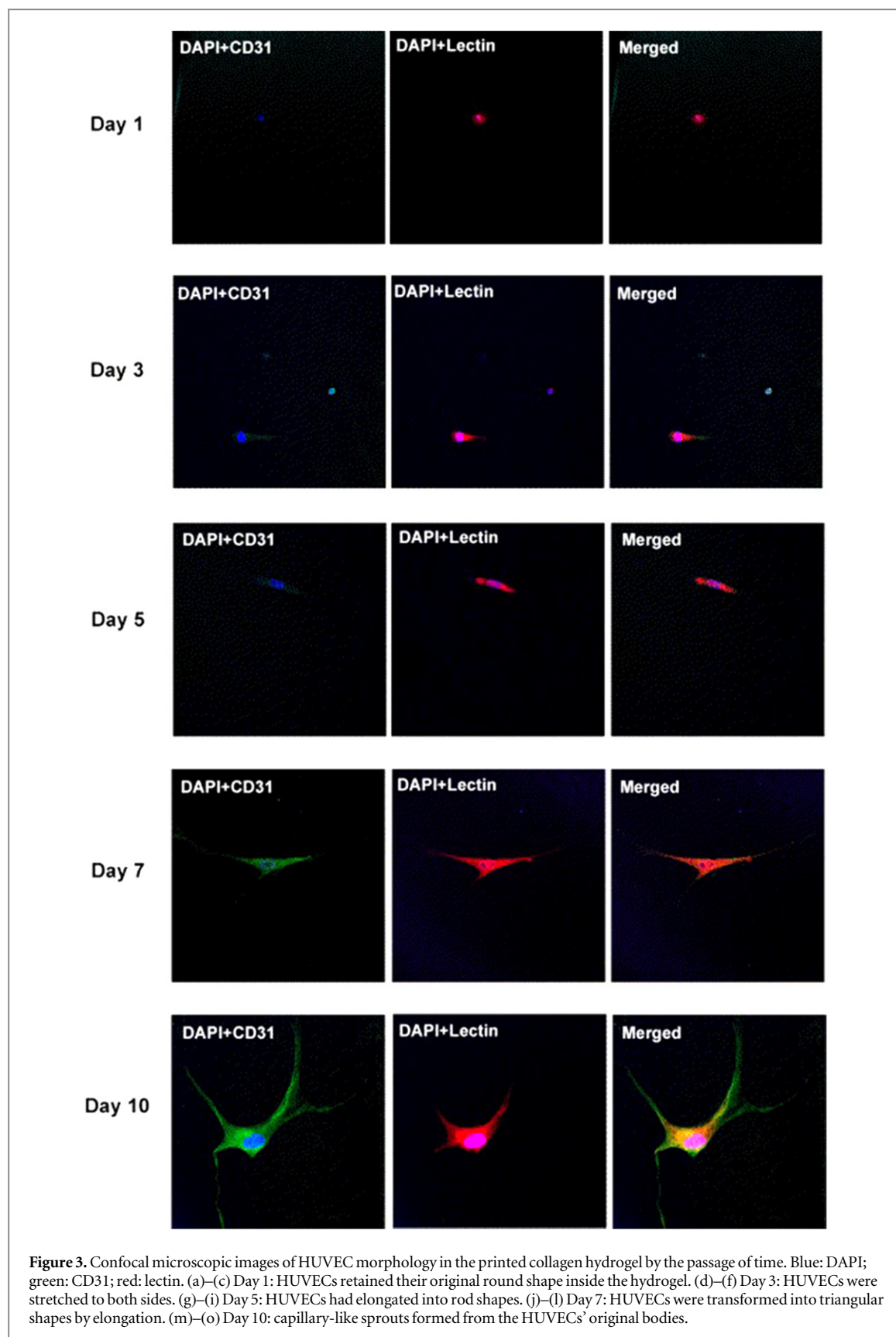
properties, and biomimicry [36]. The silk/gelatin bioink recently developed successfully fabricated a 3D cell-laden hydrogel construct, and the construct maintained its shape without PCL frameworks [37]. This bioink is likely appropriate for cell printed soft tissues, such as liver tissues.

Many researchers have tried to use micromachining and/or micro-molding/photolithographic approaches to fabricate single-layer and multi-layer constructs [38, 39]. Although a few studies have produced results showing the potential for tissue regeneration, these methods cannot control the position and density of cells inside the 3D constructs. However, these limitations were less evident in our constructs, as the cells were controllably printed using a 3D cell printing system. Furthermore, the cellular density can be modulated using our system, as we can use more than one hydrogel at a time.

3.2. Vascularization in the construct

We used a confocal microscope to observe the generation of the capillary-like network in a HUVEC/HLF co-cultured 3D cell printed construct. After printing the cell-laden collagen solution into the canals of the PCL framework, we stained the HUVECs with CD31, lectin, and DAPI to observe the morphology (figure 3) of the HUVECs in the collagen hydrogel on Days 1, 3, 5, 7, and 10 of culture time. On Day 1, the HUVECs retained their original round shape inside the hydrogel; on Day 3, they were stretched to both sides; and by Day 5, they had elongated into rods. The HUVECs were then transformed into a triangular shape by further elongation. On Day 10, capillary-like sprouts were generated from the HUVECs' original bodies and tried to link with other HUVECs. These results demonstrate that our printed hydrogel environment has the potential to support the growth of HUVECs.

To visualize the capillary-like network, we used lectin staining under culture conditions of high concentrations of HUVECs and HLFs. HUVECs and HLFs were mixed at a ratio of 1:3 in the hydrogel, and the total density of the two types of cells was set to 1×10^5 cells ml⁻¹. Images (figure 4) of the 3D capillary-like network were obtained using confocal microscopy. On Day 1, no transformation of the HUVECs was observed (figures 4(a) and (b)). On Day 3, the HUVECs started to generate capillary-like tubes (figures 4(c) and (d)). On Day 10, the HUVECs formed a capillary-like network, and most of the cells extended and created a complex capillary-like network (figures 4(e) and (f)). On Day 14, the HUVECs formed a 3D capillary-like network and filled the hydrogel structure densely (figures 4(g) and (h)). The capillary-like network formation area was calculated from the lectin staining result, which showed that the network increased approximately linearly over time (figure 5). The network occupied 4.8% of the total hydrogel area on Day 3, 11.6% on Day 7, and 36.9% on Day 14. This



complex 3D capillary-like network formation and the rapid increase in its area demonstrated that the co-culture system functioned successfully.

The above staining results show that the HUVECs developed interconnected capillary networks with

lumens within the gel environment: a collagen hydrogel containing HLFs. These results are in accordance with the fact that fibroblasts promote angiogenesis *in vitro* by secreting soluble mediators that diffuse through hydrogel to endothelial cells [40, 41].

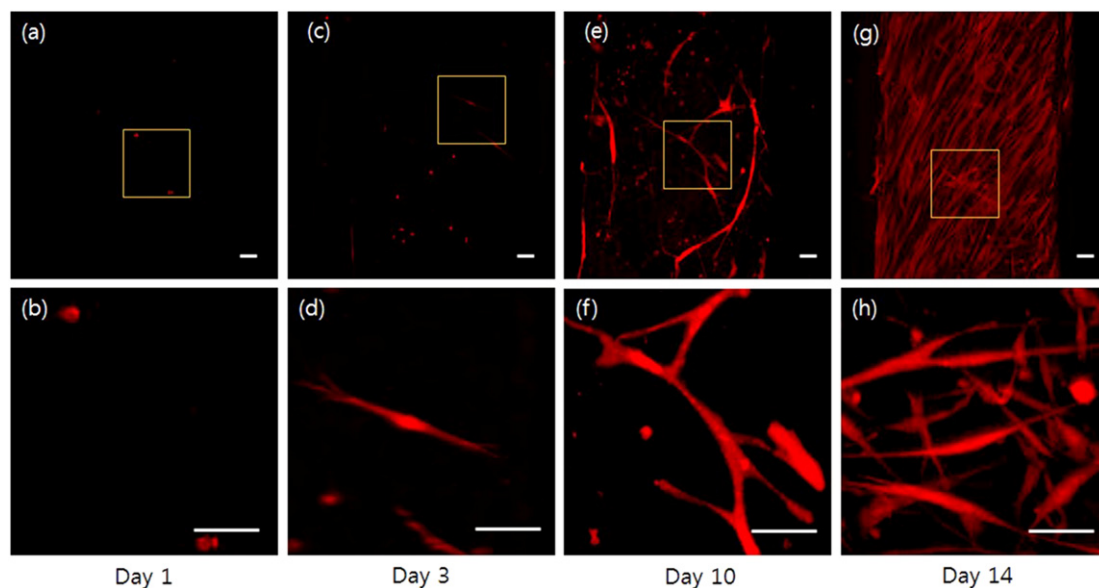


Figure 4. Capillary network formation in the printed collagen hydrogel line from confocal microscopy. (a), (b) Day 1: no transformation of the HUVECs was observed. (c), (d) Day 3: HUVECs started to generate capillary-like tube. (e), (f) Day 10: HUVECs formed a capillary network. (g), (h) Day 14: HUVECs had formed a 3D capillary network and had densely filled the hydrogel structure. (Scale bar: 50 μm .)

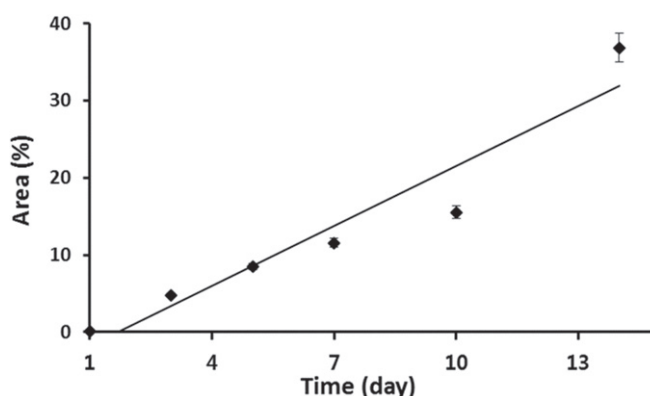


Figure 5. Change over time (x) in areas (y) covered by capillary network formation in the 3D cell printed scaffold. Areas of capillary network generated from the HUVECs increased linearly over the duration of the experiment. (Regression: $y = 2.5918x - 4.3857$, $r^2 = 0.915$, bars: $\pm$ s.d..)

Therefore, we demonstrated that our 3D co-cultured system considering angiogenesis might be an excellent microenvironment for liver tissue engineering.

3.3. Albumin secretion and urea synthesis in co-cultured 3D cell printed construct

The liver provides various essential functions, including protein synthesis and metabolism. To observe the functionality of our 3D cell printed construct for liver tissue regeneration, we measured the secretion level of albumin, which is a representative protein synthesized by HCs, and the synthesis level of urea, which is indicative of the metabolism of HCs.

To observe the albumin secretion and urea synthesis of HCs in a microenvironment with several types of cells, we printed cell-laden hydrogels under three

different conditions (table 1). After preparation of the 3D cell printed construct, we harvested the culture medium at Days 1, 4, 7, and 10 for analysis (figure 6). In Group 1, a very small amount of albumin was measured; this result shows that the viability and functionality of the cells were extremely low when only HCs were seeded in the hydrogel. In Group 2, a relatively large amount of albumin was secreted on Day 1, but the secreted amount decreased gradually afterward; this result suggests that HC viability and functionality were higher within the co-culture of HCs and HLFs than in a monoculture of HCs. However, the decline of the albumin secretion level as the culture time passed demonstrated a gradual decrease in the ability of the HCs in Group 2 to synthesize protein. In Group 3, the albumin secretion level was the highest among the three groups for the entire culture duration, and the

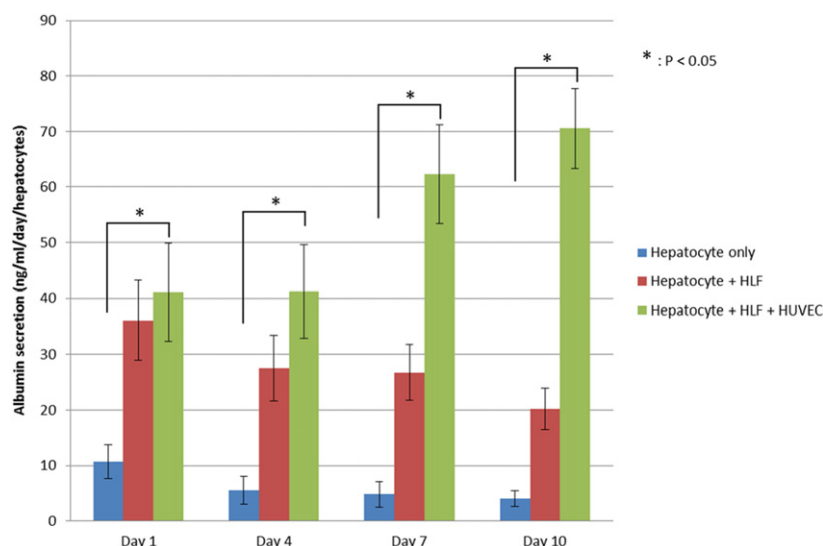


Figure 6. Albumin secretion concentration, secreted by HCs per 24 h at four sampling intervals. At each culturing time of Days 1, 4, 7 and 10, the distinction between three groups was statistically significant ($p < 0.005$). Differences in urea concentration for the entire culturing duration of Group 1 (hepatocyte only), Group 2 (hepatocyte + HLF) and Group 3 (hepatocyte + HLF + HUVEC) were also statistically significant ($p < 0.05$) (not marked in the figure). All data were normalized from the cell number of Group 3. Error bars represent s.d..

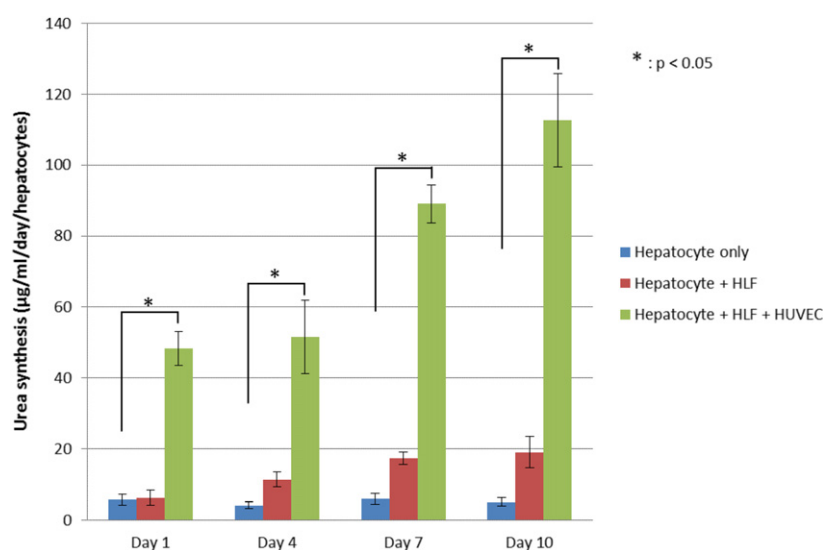
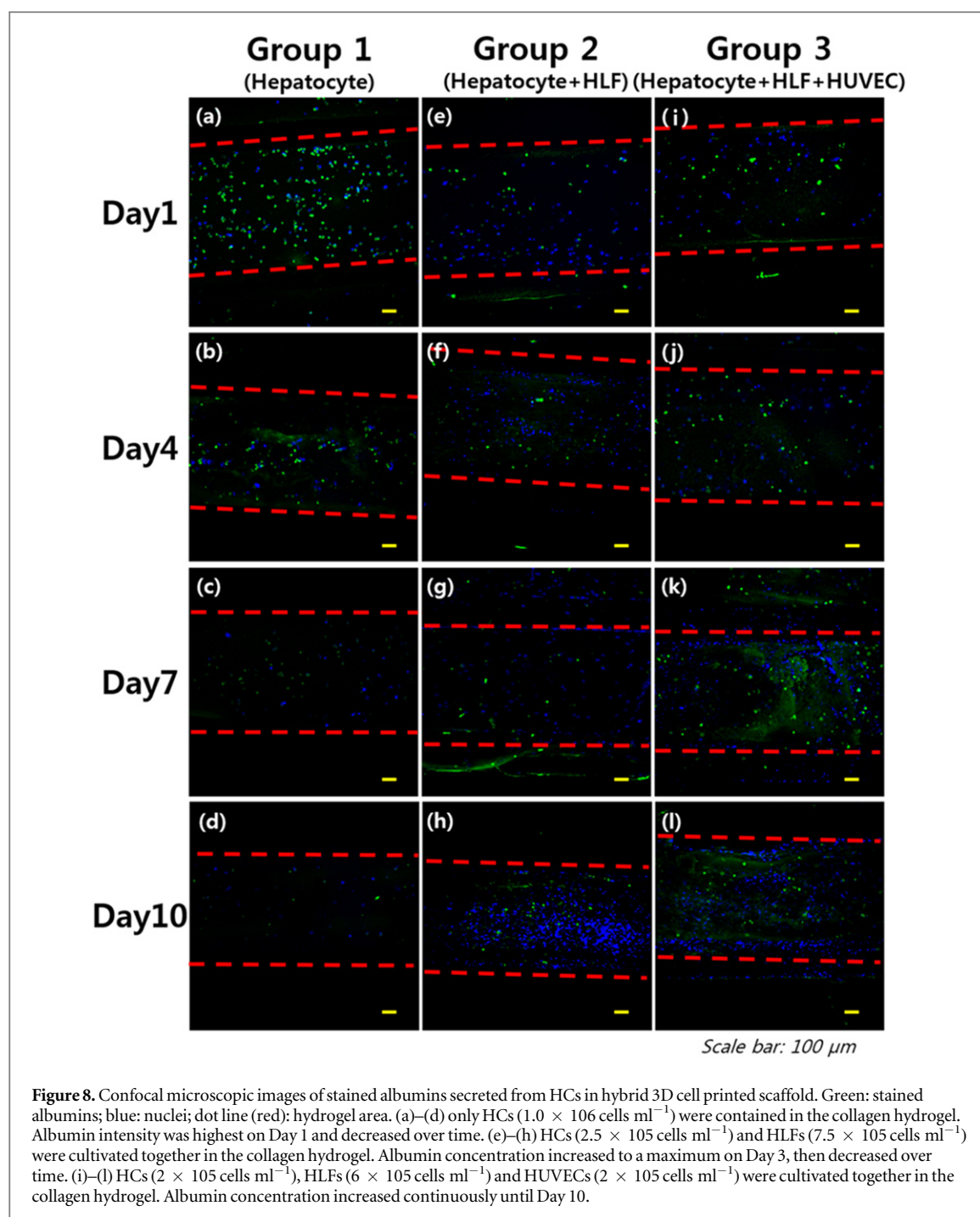


Figure 7. Urea synthesis concentration, secreted by HCs for 24 h at four sampling intervals. At each culturing time of Days 1, 4, 7 and 10, the distinction between three groups was statistically significant ($p < 0.001$). Differences in urea concentration for the entire culturing duration of group 2 (hepatocyte + HLF) and Group 3 (hepatocyte + HLF + HUVEC) were also statistically significant ($p < 0.005$) (not marked in the figure). All data were normalized from the cell number of Group 3. Error bars represent s.d..

level also increased over time. Notably, on Day 10, the albumin level in Group 3 was 15 times higher than that of Group 1 and 3.5 times higher than that of Group 2.

The urea concentration changed over time and varied among the Groups (figure 7). In Group 1, only a small amount of urea was detected; this result was similar to the albumin secretion result for the same group. In Group 2, the amount of synthesized urea increased gradually over time, but remained lower than that of Group 3 for the entire duration of the culture. In Group 3, the urea synthesis was highest and increased rapidly over time.

To evaluate the functionality of HCs further, we conducted albumin staining and observed the stained cells under a confocal microscope (figure (8)). In Group 1 (figures 8(a)–(d)), the albumin intensity was highest on Day 1 and decreased over time. In Group 2 (figures 8(e)–(h)), the albumin concentration increased to a maximum at Day 3 and then decreased over time. In Group 3 (figures 8(i)–(l)), the albumin concentration increased continuously until Day 10. These results further confirm that the highest functionality of HCs occurred when they were printed together with HUVECs and HLFs.



To overcome the limitations of 2D environments, 3D approaches to culture HCs, including hollow fibers, macroporous polymer scaffolds, cell encapsulation in hydrogels, and 3D cell cultures with hepatic cell sheets, have been investigated. These methods show the potential for liver tissue regeneration. In the 3D printing field, approaches that use biomaterials, such as gelatin, chitosan, alginate, and PEGDA, and combinations thereof have been developed [42, 43]. Furthermore, to help circumvent the limitations of the supply of nutrients and oxygen, several techniques have been attempted that include printed sacrificial molds to make a vascular network that utilizes photo-cross-linkable materials, including PEGDA gel, fibrin gel,

Matrigel, and agarose gel [29]. However, the structures developed with these techniques lack mechanical rigidity. In this study, we satisfied the requirements of both the hepatic functionality and structural stability of the scaffold.

The interaction of HCs with HUVECs and HLFs appeared to stimulate albumin production in our study, as the high albumin production levels were maintained for the entire culture time. Although the precise mechanism remains unknown, the regulation of albumin expression has been shown to be transcriptionally controlled by various transcription factors, one of which seems to cause the increase in albumin expression observed in rat HCs in co-culture

with HUVECs [44]. Fibroblasts promote angiogenesis *in vitro* in part by secreting soluble mediators that diffuse through the hydrogel to endothelial cells [40, 41]. Our experimental results are also consistent with the existing knowledge that HLF promotes the angiogenesis of HUVECs. We believe that urea synthesis might have increased for the same reason. Therefore, potential mediators of cell–cell communication signals secreted by cells could affect stabilization and increase cell functionality [45]. According to a previous study, the formation of the healthy morphologies of bile canaliculi, which were produced by hepatocytes, fibroblasts, and endothelial cells, could facilitate the effective delivery of culture medium to the construct [46, 47]. Although we have used HLF for this study, other types of fibroblasts, such as dermal fibroblasts, could be evaluated for their potential to improve the functions of HCs and induce capillary-like network formation, as these types of fibroblasts would provide easy accessibility in clinical scenarios [48–50]. Since the diffusion capacity is limited to a distance of less than 500 μm [51], supplying sufficient nutrients and oxygen to the large 3D cell-laden construct should be considered. Numerous approaches have been attempted to overcome this limitation. One of the methods involves creating a perfusable vascular network in the 3D cell-laden construct [52–54]. This method is used to print the sacrificial channels that later encapsulate within the hydrogel. The sacrificial channels are then removed, and the HUVECs can be seeded. The HUVECs later form into endothelial tubes that allow the perfusion of media. In our study, although the thickness of our 3D cell printed construct was more than 500 μm , high cell viability and functionality were observed. We believe that printing with HUVECs and vascularization of the construct might supply sufficient nutrients and oxygen to cells. In this study, we concentrated on creating simple 3D constructs with interconnected porosity; however, future studies could attempt to create liver-mimicking architecture using similar techniques based on the 3D printing method developed by our laboratory [55].

4. Conclusions

Until now, researchers have mostly utilized 2D planar environments to recreate liver tissue morphology and functions. However, these studies have been limited by their inability to replicate the 3D environment of liver tissue (i.e., the actual living environment of HCs). Here, we described and evaluated a method for using 3D cell printing technology with an MtoBS to fabricate a 3D cell printed construct for liver tissue engineering. We used PCL as the framework and collagen solution as the bioink. We encapsulated HCs, HUVECs, and HLFs in a collagen solution, and then fabricated the 3D cell printed construct by printing the cell-laden collagen into the canals of the PCL framework. Co-

culture with nonparenchymal cells was essential for HC survival; they did not survive when cultured alone *in vitro*. We also confirmed that a 3D cell printed construct containing a capillary-like network improved the protein secretion and metabolism of HCs. We believe that our 3D cell printed construct with a capillary-like network has great potential for functional liver tissue regeneration.

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